

Q] $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \dots$

$S = ?$

Solⁿ: $\frac{\frac{1}{2}}{1} = \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{\frac{1}{8}}{\frac{1}{4}} = \frac{1}{2} = r$

$\Rightarrow$ Common ratio = $r = \frac{1}{2}$

$\Rightarrow$ GP series: $S_{\infty} = \frac{a}{1-r} = \frac{1}{1-\frac{1}{2}} = 2 //$

Q] Which term of $\left(\frac{1}{4}\right), -\frac{1}{2}, 1, \dots$ is -128 ?

Solⁿ: $\frac{-1/2}{1/4} = \frac{1}{-1/2} = -2 = r$

$\Rightarrow$ GP series

$\Rightarrow T_n = ar^{n-1}$

$-128 = \frac{1}{4} (-2)^{n-1}$

$-(2^7) = \frac{1}{2^2} (-2)^{n-1}$

2	128
2	64
2	32
2	16
2	8
2	4
2	2
	1

$$128 = 2^7$$

...

$$-(2^7) = \frac{1}{2^2} (-2)^{n-1}$$

$$-2^9 = (-2)^{n-1}$$

$$(-2)^9 = (-2)^{n-1}$$

$$9 = n-1$$

$$\boxed{n = 10}$$

Q] If the product of 3 consecutive terms of GP is 27, find middle term

Solⁿ:

$$a, ar, ar^2$$

(M1)

$$a(ar)(ar^2) = 27$$

$$a^3 r^3 = 27$$

$$\boxed{ar = 3}$$

GP series (3 consecutive)

$$\frac{a}{r}, a, ar$$

(M2)

$$\Rightarrow \frac{a}{r} \cdot a \cdot ar = 27$$

$$a^3 = 27$$

$$\boxed{a = 3}$$

Q) $\underline{0.9} + 0.09 + 0.009 + \dots ?$

Solⁿ : $\frac{0.09}{0.9} = \frac{0.009}{0.09} = 0.1$

$\Rightarrow$ GP series

$\Rightarrow \boxed{r = 0.1}$

$\Rightarrow S_{\infty} = \frac{a}{1-r} = \frac{0.9}{1-0.1} = 1$

AGP series

(M1)

Q]

$$S = \frac{2}{5} + \frac{4}{5^2} + \frac{6}{5^3} + \frac{8}{5^4} + \dots$$

①

$$\frac{S}{5} = \frac{2}{5^2} + \frac{4}{5^3} + \frac{6}{5^4} + \dots$$

$$S - \frac{S}{5} = \frac{2}{5} + \left(\frac{4}{5^2} - \frac{2}{5^2} \right) + \left(\frac{6}{5^3} - \frac{4}{5^3} \right) + \left(\frac{8}{5^4} - \frac{6}{5^4} \right) + \dots$$

$$\frac{4}{5} S = \frac{2}{5} + \frac{2}{5^2} + \frac{2}{5^3} + \frac{2}{5^4} + \dots$$

$$\frac{4}{5} S = \frac{2}{5} + \frac{2}{5^2} + \frac{2}{5^3} + \frac{2}{5^4} + \dots$$

$$\frac{4S}{5} = 2 \left[\frac{1}{5} + \frac{1}{5^2} + \frac{1}{5^3} + \dots \right]$$

$$S_{\infty} = \frac{\frac{1}{5}}{1 - \frac{1}{5}}$$

$$\Rightarrow \frac{4S}{5} = 2 \times \frac{\frac{1}{5}}{\frac{4}{5}} \Rightarrow S = \frac{5}{8}$$

$$Q] \quad S = \frac{2}{5} + \frac{4}{5^2} + \frac{6}{5^3} + \frac{8}{5^4} + \dots$$

(M2)

$$S = \frac{2}{5} \left[1 + \frac{2}{5} + \frac{3}{5^2} + \frac{4}{5^3} + \dots \right]$$

$$= \frac{2}{5} \left[1 + 2\left(\frac{1}{5}\right) + 3\left(\frac{1}{5}\right)^2 + 4\left(\frac{1}{5}\right)^3 + \dots \right]$$

$$= \frac{2}{5} \left[1 - \left(\frac{1}{5}\right) \right]^{-2}$$

$$(1-x)^{-2} = \underline{1} + 2x^1 + 3x^2 + 4x^3 + \dots$$

$$= \frac{2}{5} \left[1 - \left(\frac{1}{5} \right) \right]^{-2}$$

$$= \frac{2}{5} \left[\frac{4}{5} \right]^{-2}$$

$$= \frac{2}{5} \times \frac{25}{16} \sqrt{}$$

$$= \frac{5}{8} //$$

Harmonic Progression [HP]

(AP)

$$a, a+d, a+2d, \dots, a+(n-1)d$$

HP

$$\frac{1}{a}, \frac{1}{a+d}, \frac{1}{a+2d}, \dots$$

$$\frac{1}{a+(n-1)d} \rightarrow T_n$$

$$\Rightarrow S_{HP} = \frac{1}{d} \ln \left[\frac{2a + (2n-1)d}{2a-d} \right]$$

Q] IF $\frac{1}{4}, \frac{1}{x}, \frac{1}{10}$ are in HP.

Then value of x is ?

Solⁿ :

4, x , 10 [AP series]

$$x = \frac{4+10}{2} = \frac{14}{2} = 7 //$$

Q] $\frac{1}{3}, \frac{1}{7}, \frac{1}{11}, \frac{1}{15} \dots$ Find 10th term of HP

Solⁿ: $\{3, 7, 11, 15, \dots\}$ AP

$$\Rightarrow d = 4$$

$$\Rightarrow a = 3$$

$$\Rightarrow T_n = a + (n-1)d$$

$$T_n = 3 + (n-1)4$$

$$T_n = 3 + (n-1)4$$

$$T_{10} = 3 + (10-1)4$$

$$= 3 + (9)4$$

AP

$$T_{10} = 39$$

HP series $\Rightarrow$ 10th term = $\frac{1}{39}$ //

Special

$$\textcircled{1} \quad 1 + 1 + 1 + \dots \quad n \text{ times} = n$$

$$\begin{aligned} \textcircled{2} \quad & 1 + 2 + 3 + \dots + n \\ &= \text{Sum of 1st } n \text{ natural number} \\ &= \frac{n(n+1)}{2} \end{aligned}$$

$$\therefore \textcircled{3} \quad 1^2 + 2^2 + 3^2 + \dots + n^2$$

$$= \frac{n(n+1)(2n+1)}{6}$$

$$\textcircled{4} \quad 1^3 + 2^3 + 3^3 + \dots + n^3$$

$$= \left[\frac{n(n+1)}{2} \right]^2 = \frac{n^2(n+1)^2}{4} //$$

$$9) \quad 10^2 + 11^2 + 12^2 + \dots + 19^2 = ?$$

$$\text{Soln: } S_{19} = 1^2 + 2^2 + 3^2 + \dots + 9^2 + 10^2 + \dots + 19^2$$

$$S_9 = 1^2 + 2^2 + 3^2 + \dots + 9^2$$

$$Req = S_{19} - S_9$$

$$= \frac{19(19+1)(2 \times 19 + 1)}{6} - \frac{9(9+1)(2 \times 9 + 1)}{6}$$

$$= 2185 //$$

8] $\frac{1}{2 \times 4} + \frac{1}{4 \times 6} + \frac{1}{6 \times 8} + \dots + \frac{1}{18 \times 20}$

$$\frac{1}{2 \times 4} = \frac{1}{2} \left[\frac{1}{2} - \frac{1}{4} \right]$$

$$\frac{1}{4 \times 6} = \frac{1}{2} \left[\frac{1}{4} - \frac{1}{6} \right]$$

$$\frac{1}{6 \times 8} = \frac{1}{2} \left[\frac{1}{6} - \frac{1}{8} \right]$$

$$\Rightarrow \left(\frac{1}{2}\right) \left[\frac{1}{2} - \frac{1}{4}\right] + \left(\frac{1}{2}\right) \left[\frac{1}{4} - \frac{1}{6}\right] + \left(\frac{1}{2}\right) \left[\frac{1}{6} - \frac{1}{8}\right] + \dots + \left(\frac{1}{2}\right) \left[\frac{1}{18} - \frac{1}{20}\right]$$

$$\Rightarrow \frac{1}{2} \left[\overset{\checkmark}{\frac{1}{2}} - \cancel{\frac{1}{4}} + \cancel{\frac{1}{4}} - \cancel{\frac{1}{6}} + \cancel{\frac{1}{6}} - \cancel{\frac{1}{8}} + \dots - \cancel{\frac{1}{18}} \overset{\checkmark}{\frac{1}{20}} \right]$$

$$\Rightarrow \frac{1}{2} \left[\frac{1}{2} - \frac{1}{20} \right] = \frac{9}{40} //$$

Q] [hw]

$$\frac{1}{2 \times 5} + \frac{1}{5 \times 8} + \frac{1}{8 \times 11} + \dots + \frac{1}{32 \times 35}$$

$$Ans = \frac{11}{70}$$

Q] 1, 4, 9, 16, 25, —

Soln:

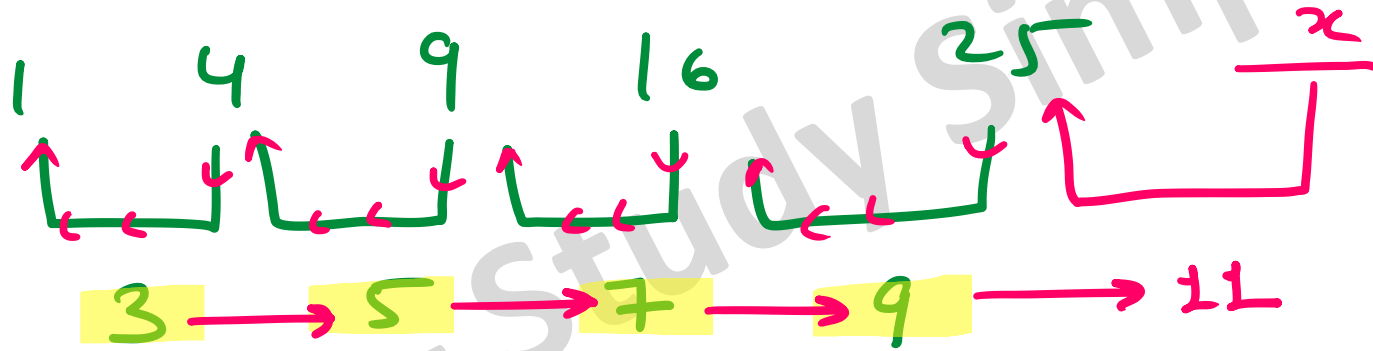
Square of natural no.

(M1)

$1^2, 2^2, 3^2, 4^2, 5^2, (6^2 = 36)$

1, 4, 9, 16, 25, —

(M2)



$$x - 25 = 11$$

$$\begin{aligned} x &= 11 + 25 \\ &= 36 \end{aligned}$$

∴ [ESE 2018]

The sum of squares of successive integers 8 to 16 both inclusive will be

(A) 1126

(B) 1174

(C) 1292

(D) 1356

$$\therefore R_{eq} = 8^2 + 9^2 + \dots + 16^2$$

$$= S_{16} - S_7$$

$$= \frac{16 \times 17 \times 33}{6} - \frac{7 \times 8 \times 15}{6}$$

$$= 1496 - 140$$

$$= 1356 //$$

[2018 EC]

$$1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \dots$$

(A) 2

(B) $7/4$

(C) $3/2$

(D) $4/3$

$\Rightarrow$ GP series

$$\Rightarrow r = 1/4$$

$$\Rightarrow S_{\infty} = \frac{a}{1-r}$$

$$= \frac{1}{1-1/4} = \frac{4}{3}$$

[GATE]

For $x = \frac{\pi}{6}$, sum of $\sum_{n=1}^{\infty} (\cos x)^{2n}$
 $= \cos^2 x + \cos^4 x + \dots$

(A) π

(B) 3

(C) ∞

(D) 1

$$\cos^2 x + \cos^4 x + \cos^6 x + \dots$$

$\Rightarrow$ GP series

$$\Rightarrow r = \cos^2 x$$

$$\Rightarrow r < 1$$

$$\Rightarrow S_{\infty} = \frac{a}{1-r} = \frac{\cos^2 x}{1 - \cos^2 x} = \frac{\cos^2 x}{\sin^2 x} = \cot^2 x$$

$$S = \omega t^2 \kappa$$

$$S\left(x = \frac{\pi}{6}\right) = \cot^2\left(\frac{\pi}{6}\right)$$

$$= \frac{1}{\tan^2\left(\frac{\pi}{6}\right)}$$

$$= \frac{1}{\left(\frac{1}{\sqrt{3}}\right)^2} = 3 //$$

①

a, b, c

(i) AP → Arithmetic Mean $\Rightarrow b = \frac{a+c}{2}$

(ii) GP → Geometric Mean $\Rightarrow b = (ac)^{1/2}$

(iii) HP → Harmonic Mean $\Rightarrow$ $b \neq \frac{a+c}{2}$

$$b = \frac{2ac}{a+c} //$$

$$a, b, c \rightarrow HP$$

$$\boxed{\frac{1}{a}, \frac{1}{b}, \frac{1}{c}} \rightarrow AP$$

$$\frac{1}{b} - \frac{1}{a} = \frac{1}{c} - \frac{1}{b}$$

$$\left(\frac{1}{b} + \frac{1}{b} \right) = \frac{1}{c} + \frac{1}{a}$$

$$\frac{2}{b} = \frac{a+c}{ac}$$

$$\Rightarrow \boxed{b = \frac{2ac}{a+c}}$$

$$AM \cdot HM = \left(\frac{a+c}{2} \right) \cdot \frac{2ac}{(a+c)}$$

$$(AM)(HM) = ac$$

$$(GM) = (ac)^{1/2}$$

$\Rightarrow$

$$(AM) \cdot (HM) = (GM)^2$$

special
case

a, b, c

$$\{a=b=c\}$$

(i) AP $\rightarrow$ Arithmetic Mean $\Rightarrow b = \frac{a+c}{2} = k$

(ii) GP $\rightarrow$ Geometric Mean $\Rightarrow b = (ac)^{1/2} = k$

(iii) HP $\rightarrow$ Harmonic Mean $\Rightarrow$ $b \neq \frac{a+c}{2}$

$$b = \frac{2ac}{a+c} = k$$

$$AM = GM = HM$$

$$AM > GM > HM$$

Shrenik Jain — Study Simplified